Neuroscience 289 (2015) 334–348

MATHEMATICALLY GIFTED ADOLESCENTS MOBILIZE ENHANCED WORKSPACE CONFIGURATION OF THETA CORTICAL NETWORK DURING DEDUCTIVE REASONING

L. ZHANG,[a] J. Q. GAN[a,b] AND H. WANG[a]*

[a] *Key Lab of Child Development and Learning Science of Ministry of Education, Research Center for Learning Science, Southeast University, Nanjing, Jiangsu 210096, China*

[b] *School of Computer Science and Electronic Engineering, University of Essex, Colchester, UK*

Abstract—Previous studies have established the importance of the fronto-parietal brain network in the information processing of reasoning. At the level of cortical source analysis, this eletroencepalogram (EEG) study investigates the functional reorganization of the theta-band (4–8 Hz) neurocognitive network of mathematically gifted adolescents during deductive reasoning. Depending on the dense increase of long-range phase synchronizations in the reasoning process, math-gifted adolescents show more significant adaptive reorganization and enhanced ''workspace'' configuration in the theta network as compared with average-ability control subjects. The salient areas are mainly located in the anterior cortical vertices of the fronto-parietal network. Further correlation analyses have shown that the enhanced workspace configuration with respect to the global topological metrics of the theta network in math-gifted subjects is correlated with the intensive frontal midline theta (fm theta) response that is related to strong neural effort for cognitive events. These results suggest that by investing more cognitive resources math-gifted adolescents temporally mobilize an enhanced task-related global neuronal workspace, which is manifested as a highly integrated fronto-parietal information processing network during the reasoning process. **© 2015 IBRO. Published by Elsevier Ltd. All rights reserved.**

Key words: math-gifted adolescents, functional network reorganization, frontal midline theta response, cortical source analysis, graph theory, workspace configuration.

INTRODUCTION

One of the most exceptional traits of mathematically gifted children/adolescents is the capacity of solving long-chain reasoning problems (Banfield, 2005). As stated in the parieto-frontal integration theory (P-FIT) that emphasizes the crucial functional interaction among discrete brain regions in reasoning processes (Jung and Haier, 2007), the previous neuroscience studies have demonstrated that the superior reasoning ability of math-gifted children/adolescents is supported by an enhanced fronto-parietal network, which is primarily composed of the structurally heightened interhemispheric interaction through the corpus callosum (CC), enhanced intrahemispheric fronto-parietal connectivity, and interhemispheric frontal connectivity between the dorsolateral prefrontal cortex (DLPFC) and premotor cortex (O'Boyle et al., 2005; Prescott et al., 2010; Navas-Sanchez et al., 2013). Besides, math-gifted children/adolescents show functionally widespread activations and additional involvements of the fronto-parietal brain regions when performing fluid reasoning tasks (O'Boyle et al., 2005; Prescott et al., 2010; Desco et al., 2011). Moreover, neuroimaging studies have consistently reported that the prefrontal cortex (PFC), frontal cortex, and anterior cingulate (AC) of the math-gifted brain show intensive event-related activation (O'Boyle et al., 2005; Desco et al., 2011). The connectivity study of the anatomical brain network has demonstrated that the anterior sub-network including the DLPFC, premotor regions, and AC of the math-gifted brain shows higher effective connectivity, which indicates an integrated role of the anterior neural system in the executive function (Prescott et al., 2010). These findings have been suggested as the important neural mechanism of the math-gifted brain.

On the other hand, cognitive functions depend on adaptive self-organization of large-scale neuronal assemblies, which can be quantified by the changes of functional network parameters caused by the task-related spatial redistribution of connections, i.e., adaptive network reconfiguration. The dynamic network reorganization is greatly related to the emergence of long-range connections (Bassett et al., 2006; Kitzbichler et al., 2011), which may play a prominent role in large-scale communication and information integration (Markov et al., 2013). Among distributed neurons, long-distance connectivity can provide a ''global workspace'' that potentially interconnects multiple distributed and

*Corresponding author. Address: Research Center for Learning Science, Southeast University, 2 Sipailou Road, Nanjing, Jiangsu 210096, China.
E-mail address: hxwang@seu.edu.cn (H. Wang).

Abbreviations: AC, anterior cingulate; ANOVA, analysis of variance; DLPFC, dorsolateral prefrontal cortex; EEG, eletroencepalogram; EOG, electrooculographic; ERPCOH, event-related phase cross-coherence; ERSP, event-related spectral perturbation; PFC, prefrontal cortex; PLV, phase-locking value; PSD, power spectrum density; RAPM, Raven's Advanced Progressive Matrices.

http://dx.doi.org/10.1016/j.neuroscience.2014.12.072
0306-4522/© 2015 IBRO. Published by Elsevier Ltd. All rights reserved.

specialized brain areas in a coordinated manner (Dehaene et al., 1998). On the other side, the intense mobilization/driving of a global workspace might be associated with individual cognitive effort (Dehaene and Naccache, 2001; Kitzbichler et al., 2011), which is defined as the total amount of cognitive resources needed to complete a task, including perception, memory, judgment, etc. (Cooper-Martin, 1994). As it is correlated with long-distance connectivity, the workspace formation can be theoretically measured by a topological shift when human brain functional networks transiently adopt a globally synchronized and highly integrated configuration pattern, i.e., workspace configuration, to replace the original community structure composed of locally synchronized and modular subsystems of unconscious processing (Dehaene and Changeux, 2005; Kitzbichler et al., 2011).

Based on the above evidence, this eletroencepalogram (EEG) study aims to investigate the capacity of dynamic network reorganization of math-gifted adolescents in performing a deductive reasoning task. The following hypotheses are proposed in this study: As compared with non-gifted subjects, math-gifted adolescents might show higher temporal integration among discrete brain regions through more long-distance fronto-parietal connections emerged in the reasoning process. Thus, greater topological reorganization toward stronger workspace configuration of the functional brain network will be measured in math-gifted subjects, which are expected to be the consequence of higher mental effort invested in the reasoning process.

This study focuses on theta-band cortical signals because of its specificity in the dynamic networking involved in the executive process of cognitive tasks (Sauseng et al., 2005, 2010; Mizuhara and Yamaguchi, 2007). Event-related theta oscillations are related to cognitive and memory performance (Klimesch, 1999). Specifically, theta synchronization might play an important role in the encoding of new information and the integration of different cognitive sub-processes in a working memory system (Klimesch, 1999; Sauseng et al., 2010). In the selectively distributed theta system of the brain, ''frontal theta'' has a ''response-controlling'' function and its responsiveness has been interpreted as a functional indication of the hippocampo-fronto-parietal neural connection system in cognitive processes (Basar et al., 2001). The previous studies have reported that theta fronto-parietal long-range phase synchronizations dynamically coordinate the central executive circuit of the prefrontal cortex and relevant cortical regions (Sauseng et al., 2005; Mizuhara and Yamaguchi, 2007). The precise fine tuning of the theta fronto-parietal network has been suggested as a reflection of the varying central executive demands (Sauseng et al., 2005). Based on the evidence from the previous studies and a time–frequency analysis of the deductive reasoning task in this study, the proposed hypotheses are tested in the following manner: (1) theta-band functional networks in the baseline and task periods are constructed from the cortical source currents; (2) the group difference in network reorganization is quantified by the changes of key graph-theoretical measures involved from locally modular community structure in resting state to globally synchronized network organization in the reasoning process, primarily including the reduced minimum path length or increased global efficiency, the diminished local cluster or local efficiency, and the decreased functional segmentation or modularity, etc. (Kitzbichler et al., 2011); (3) while the workspace pattern of the theta network has been formed in the reasoning task, the group effect on workspace configurations is assessed by the topological property of globally integrated network architecture that is associated with less modular and more efficient network configuration; (4) by using task-induced ''fm theta'' response (i.e., the theta oscillation in frontal midline brain area) as the indicator of the effortful executive manipulation on the cognitive workload (Turken et al., 1999; Onton et al., 2005; Doppelmayr et al., 2008), the effort-driven topological reorganization and workspace formation of the theta network are estimated by single-trial correlation analyses, in which the giftedness-related neural features are distinguished whereby the relevant psychological mechanisms of math-gifted adolescents are analyzed and discussed.

EXPERIMENTAL PROCEDURES

Subjects

This study enrolled 24 adolescent subjects without left handedness, medical, neurological, or psychiatric illness, and history of brain injury or surgery. The study was approved by the Academic Committee of the Research Center for Learning Science, Southeast University, China. All the subjects were asked to read and sign a fully informed consent form for this experiment and received financial compensation for their participation.

The math-gifted group included 11 students (eight males and three females) aged 15–18 years (mean ± SD: 16.3 ± 0.6) from the Science and Engineering Experimental Class at Southeast University, Nanjing, China. The class was composed of gifted adolescents with exceptional abilities in mathematics and natural sciences, who had been selected nationwide before the age of 15 years by a special college entrance examination. According to the identification and assessment of ''mathematical giftedness'' from a wide range of theories and concepts (Gagne, 1985; Gardner, 2006), the screening procedure of the math-gifted subjects includes three crucial criteria: (1) teachers' nomination: it comes from performance-based observation and judgement on mathematical giftedness; (2) mathematical ability and academic performance: they must have been awarded prizes in at least one of the nationwide or provincial mathematical competitions (e.g., Maths Olympic Competition) to demonstrate the exceptional abilities in logic thinking, visuospatial imagery, and creative thinking etc.; (3) Intelligence level: their scores on Raven's Advanced Progressive Matrices (RAPM) must be higher than 32.

The control group was composed of 13 students who were recruited from the Fourth Nanjing High School, with average-level mathematical performance in class tests.

They matched with the math-gifted group for age (mean ± SD: 15.9 ± 0.7) and gender (eight males and five females), and their intelligence scores of the RAPM test were lower than 32.

Experiment paradigm

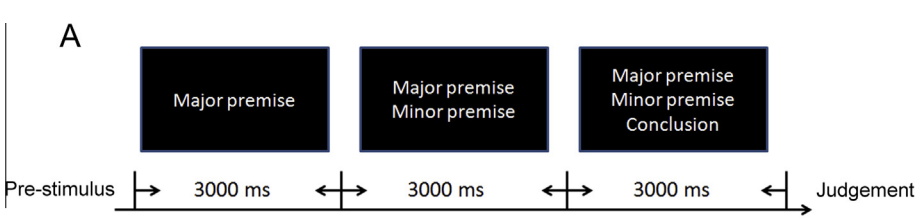
Superior logical thinking ability is regarded as a major characteristic of math-gifted students (Gardner, 2006), which is mainly constituted by the "analytic" type in a verbal–logical way and the "geometric" type in a visual–pictorial way (Krutetskii, 1976). So far, most studies have investigated the neural substrates of the math-gifted brain by using visuospatial tasks, such as RAPM and mental rotation (O'Boyle et al., 2005; Prescott et al., 2010; Desco et al., 2011). To assess the neural correlations with analytic logic of math-gifted adolescents, a verbal–logical deductive reasoning task with a type of categorical syllogism was performed by the subjects in this experiment.

As an essential skill in mathematics, deductive reasoning is a logical thinking process that draws a conclusion from given premises by following rigid deduction steps, which has been widely used in the studies on reasoning of the human brain (Wharton and Grafman, 1998; Goel, 2007; Prado et al., 2011). The categorical syllogism used in this study is composed of three basic letter items, e.g., "S", "M", and "P". A sample of the categorical syllogism is "some S are M, all M are P, therefore some S are P". Here, "S" and "M" constitute the major premise, and "M" and "P" form the minor premise, where "M" is the medium item that is presented in the two premises (Fig. 1B). The inferred relationship between "S" and "P" constitutes the conclusion of the deduction process. Corresponding to the three sentences, the deductive reasoning process is divided into three stages by most psychological studies, i.e., premise encoding, premise integration, and conclusion validation. Particularly, the brain regions activated during the latter two stages are suggested to be related to the emergence of the actual reasoning process (Rodriguez-Moreno and Hirsch, 2009).

The three-item-series categorical syllogism problem without concrete content has been found to activate the left-lateral fronto-parietal system, and also require highly imaginative situation conducted by the right hemisphere (Wharton and Grafman, 1998; Prado et al., 2011). Specifically, the left superior frontal, right medial frontal gyrus, and bilateral parietal lobule have been suggested as the "core regions" of deductive reasoning, which constitute a cooperative fronto-parietal network for task-relevant information processing. In this network, the left-lateral frontal area responds to the cognitive control and decision-making processes, the right frontal cortex is involved in mental image of the relationship between the premises, the bilateral posterior system mediates the visual premise processing and attention resource distribution, and the interplay of frontal–parietal cortices conducts hypothesis test of various solutions to the problem (Knauff et al., 2002; Fangmeier et al., 2006; Goel, 2007; Monti et al., 2007).

The experimental paradigm is constituted by a valid block (32 trials), an invalid block (32 trials), and a baseline block (40 trials). A valid trial is composed of the combination of syllogistic sentences following true logic rules. The invalid combination of syllogistic sentences results in an invalid trial, in which there is inconclusive relationship between the two premises or incorrect conclusion under clear premises. A baseline trial includes the same letter items in each sentence, which only match with the timeline of the valid/invalid trials. The sentences of each trial are composed of Chinese characters and English letters, which are white with black background to avoid visual fatigue. The letters in the sentences are randomly selected from the 26 letters of the English alphabet.

The major premise, minor premise and conclusion of each trial were presented sequentially along the timeline (Fig. 1A). The time length of each presentation is 3000 ms. So, the total presentation time of a reasoning process is 9000 ms. At the preparation stage, subjects put their left index finger on key "D" and right index finger on key "K". After the presentation of the major premise and minor premise, subjects were asked to draw a conclusion about the logic relationship between items "S" and "P". When the given conclusion was presented, subjects should judge whether the conclusion was consistent or inconsistent to their own inference. Within the limited period of 3000 ms, they were asked to respond as quickly and accurately as possible by pressing "D" for "consistent" or "K" for "inconsistent".



B

	Major premise	Minor premise	Conclusion
Valid sample	No A is B	All X are B	No A is X
Invalid sample	Some D are E	All K are D	Some K are E
Baseline sample	Some C are C	All D are D	No E is E

Fig. 1. Experiment protocol: (A) timeline of the stimulus presentation. (B) Some samples of valid, invalid and baseline trials of categorical syllogism.

The time length of the experimental task for each subject was about 30 min. All the trials of the three blocks were presented randomly in the E-Prime 2.0 procedure. After conducting a practice session composed of five trials, subjects could decide to whether to practice again or start the formal experiment.

EEG signal recording and preprocessing

The EEG data were recorded by the Neuroscan international 10–20 system with a sampling rate of 1000 Hz (Fig. 2A). Except for 60 data channels, two reference electrodes were placed on the bilateral mastoids of the subjects, and four surface electrodes simultaneously recording electrooculographic (EOG) signals were used to monitor ocular movements and eye blinks, with one pair placed over the higher and lower left eyelids and the other pair placed 1 cm lateral to the outer corner of the left and right orbits.

The continuous EEG data were preprocessed by the Scan 4.3 software. The raw signals were band-pass filtered with a frequency band of 1–60 Hz. Each trial was extracted with the epoch of 9500 ms (500-ms pre-stimulus and 9000-ms post-stimulus). The baseline-correction was performed following the pre-stimulus time interval and the ocular artifacts were removed according to the simultaneously recorded EOG signals. After the artifact rejection with the thresholds ranging from 50 μv to 75 μv, the trials contaminated by eye blinks and electrocardiogram noise were excluded. As a result, 215 task trials were retained for the math-gifted group and 243 task trials were retained for the control group, in which each subject had 18–25 trials. For further removing possible artifacts, the independent component analysis (ICA) of the EEGLAB toolbox was used to clear visible artifacts, such as the components of possible ocular and muscle movements.

Time–frequency analysis of the deductive reasoning task

In each group, the trials were concatenated to constitute a three-dimensional dataset, whose size was (No. of electrodes) × (No. of time points) × (No. of trials). The electrode based time–frequency analyses were performed to determine the intervals with significant task-induced scalp response. Bootstrapping statistical test was conducted to constrain the following analytical results to significance level of 0.05 (Delorme and Makeig, 2004).

Event-related spectral perturbation (ERSP) measures the event-related changes of a signal by averaging the spectral power values in short sliding time windows with multiple stimulus-locked trials. For observing the task-induced spectral response that varied with the three-stage reasoning process, the ERSP of electrode FZ in frontal midline location was calculated.

Event-related phase cross-coherence (ERPCOH) discovers the emergence of cortical connection over the course of a cognitive task by measuring the degree of synchronization between any two electrodes. Since the long-range coherence is crucial for functional network reorganization and workspace formation, the ERPCOH calculation was respectively conducted for electrode pair F1–F8 located at bilateral inferior frontal cortices and electrode pair AF3–PO3 at the left-lateral frontal pole-parietooccipital cortex, which aimed to find the transient shifts of interhemispheric frontal coherence and intrahemispheric frontal-parietal coherence during the deductive reasoning process.

Theta-band scalp activation and cortical source estimation

Task-induced theta response. The EEG signals were further filtered within the theta frequency band of the brain activity. The task-induced spectral response of each signal was estimated by calculating the event-related absolute power change, which was expressed as a percentage of power change in the task period with respect to the resting state (Pfurtscheller and Silva, 1999):

$$\Delta P(f) = [|A(f) - R(f)|/R(f)] \times 100\% \quad (1)$$

where f indicates the frequency band, $A(f)$ is the power spectrum density (PSD) of a signal in the task period and $R(f)$ is the PSD in the pre-stimulus interval.

Cortical source estimation. In practice, phase coherence between the scalp-recorded EEG signals was influenced by the volume conduction, which might lead to spurious estimates on functional connectivity of the brain network (Langer et al., 2012; Klados et al., 2013). To address the problem, the EEG signals were

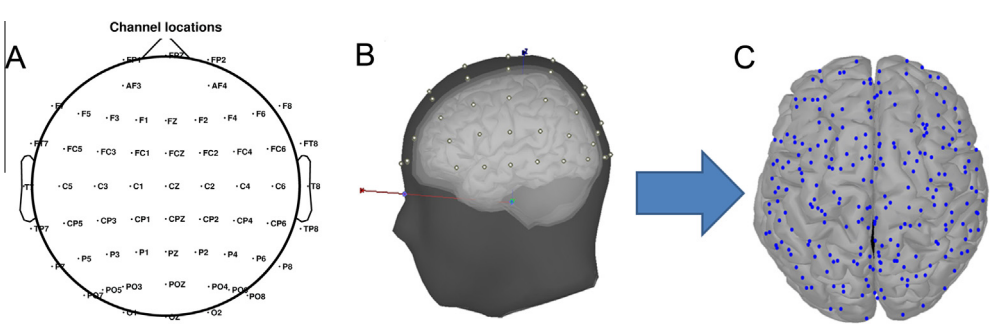


Fig. 2. Head model and cortical vertices: (A) EEG electrode placement of the Neuroscan international 10–20 system. (B) Head model with four layers (scalp, outer skull, inner skull and cortex). (C) Cortical vertices transformed from the EEG channel locations through a source estimation procedure.

transformed into the source space by using a source estimation procedure given in the Brainstorm toolbox (http://neuroimage.usc.edu/brainstorm) (Tadel et al., 2011).

In the forward process of the source estimation, the EEG signals were assumed to be primarily determined by a block of electric dipoles located at the surface of the cortex. An averaged realistic head model constituted by scalp, outer skull, inner skull, and cortex was used as the common brain for group analysis, and the symmetric Boundary Element Method (BEM) in the open-source software (http://www-sop.inria.fr/athena/software/OpenMEEG/) (Gramfort et al., 2010) was applied to the EEG electrode locations of the head model to obtain the volume conductor modeling of the subjects (i.e., the forward model matrix). The noise of the sensors was removed by computing the noise covariance matrix from the pre-stimulation time interval. In the inverse process, an inverse kernel matrix was produced by the forward model and the standardized Low Resolution Brain Electromagnetic Tomography (sLORETA), by which the EEG signals could be transformed into the cortical current sources. After further downsampling the original sources, 248 cortical vertices were selected to serve as the nodes in the following network analysis (Fig. 2B, C).

Graph-theoretical analysis of the phase synchronization network

Association matrix of phase synchronization. As the stage of premise integration in the deductive reasoning (i.e., the presentation of minor premise) was suggested as the period of the occurrence of actual reasoning, the epoch of 3000–6000 ms was further extracted for the connectivity analysis of the functional network.

Phase-locking value (PLV) between pairwise cortical currents was calculated to construct the association matrix (Lachaux et al., 1999). In the definition of PLV, $x(t)$ and $y(t)$ represent two cortical signals with equal time length, $\phi_x(t)$ and $\phi_y(t)$ are the instantaneous phases of $x(t)$ and $y(t)$, and PLV refers to the locking of the two phase signals, i.e., $|\phi_x(t) - \phi_y(t)| = const$. Here, the Hilbert transform (HT) is used to obtain the phase of a signal by constructing an analytical signal $H(t) = x(t) + i\tilde{x}(t)$, where $\tilde{x}(t)$ represents the HT of and is obtained by $\tilde{x}(t) = \frac{1}{\pi} PV \int_{-\infty}^{\infty} \frac{x(t')}{t-t'} dt'$ (PV is the Cauchy principal value). The analytical signal phase can be given by $\phi(t) = \arctan \frac{\tilde{x}(t)}{x(t)}$. Therefore, PLV bivariate metric is obtained according to

$$\mathrm{PLV} = \left| \frac{1}{M} \sum_{j=0}^{M-1} \exp(i(\phi_x(j\Delta t) - \phi_y(j\Delta t))) \right| \tag{2}$$

where Δt is the sampling period and M is the number of sample points of each signal. The value of PLV ranges from 0 to 1, where 1 denotes complete phase synchronization and 0 represents absence of synchronization (Sakkalis, 2011).

Undirected graph construction. After computing the PLV matrix with a size of 248×248 from all the pairwise nodes, an adjacency matrix ($a_{i,j}$) is obtained by using one of the following two criteria: (1) a fixed connection density is set according to the Erdös–Rényi model (Erdös and Rényi, 1961) (i.e., $p = 2 \ln n/n$, where n is the number of the nodes), which is used to find the topological differences between the task and baseline periods that are caused by task-related connection redistribution in the network; (2) a fixed threshold is set for the PLV matrices in the task period, which is the mean value of the minimum PLVs in the resting-state adjacency matrices with fixed connection density from criterion (1), to find the between-groups difference in the reasoning-related network topology. After that, graph theory is applied to the adjacency matrix to quantify the topological property of the brain functional network.

In the following definitions in the graph theory (Bullmore and Sporns, 2009; Rubinov and Sporns, 2010), N is the set of all the nodes in a network, and (i, j) represents the link between nodes i and j ($i, j \in N$). If there is connection status between the two nodes, $a_{i,j} = 1$; otherwise, $a_{i,j} = 0$.

As the most fundamental network measure, node degree is quantified by the number of connections linked to a node as follows:

$$k_i = \sum_{j \in N} a_{ij} \tag{3}$$

In a cluster within a network, node clustering coefficient describes the interaction between neighboring nodes, which is defined as the ratio of the number of existing connections between the nearest neighbors of a node to the maximum number of possible connections:

$$C_i = \frac{2t_i}{k_i(k_i - 1)}, \qquad (C_i = 0 \quad \text{if} \quad k_i < 2) \tag{4}$$

where t_i is the number of triangles around node i, and is defined as $t_i = \frac{1}{2} \sum_{j,h \in N} a_{ij} a_{ih} a_{jh}$.

Local efficiency of a network is associated with local clustering, information transfer, and robustness, which is defined as

$$E_{loc} = \frac{1}{n} \sum_{i \in N} E_{loc,i} = \frac{1}{n} \sum_{i \in N} \frac{\sum_{j,h \in N, j \neq i} a_{ij} a_{ih} \left[d_{jh}(N_i) \right]^{-1}}{k_i(k_i - 1)} \tag{5}$$

where $E_{loc,i}$ is the local efficiency of node i and $d_{jh}(N_i)$ is the length of the shortest path between nodes j and h, which contains only neighbors of node i.

Within-module degree z-score of a node describes how well the node is connected to other nodes in its (functional) module, with high value reflecting a high level of intramodule connectivity of the node, which is defined as

$$Z_i = \frac{k_i(m_i) - \bar{k}(m_i)}{\sigma^{k(m_i)}} \tag{6}$$

where m_i is the module containing node i, $k_i(m_i)$ is the within-module degree of node i (the number of links between node i and all other nodes in m_i), and $\bar{k}(m_i)$ and $\sigma^{k(m_i)}$ are the respective mean and standard deviation of the within-module degrees over all nodes in m_i.

For the global topology of a network, characteristic path length is the average number of connections in the

shortest paths between all the nodes, which is quantified by

$$L = \frac{1}{n}\sum_{i\in N} L_i = \frac{1}{n}\sum_{i\in N}\frac{\sum_{j\in N, j\neq i} d_{ij}}{n-1} \tag{7}$$

where L_i is the average distance between node i and other nodes, and $d_{ij} = \sum_{a_{uv}\in g_{i\to j}} a_{uv}$, which indicates the shortest path length between nodes i and j ($g_{i\to j}$ is the shortest geodesic path between i and j. For all the disconnected node pairs i and j, $d_{ij} = \infty$).

To measure the strength of division of a network into modules, modularity of the network is computed according to

$$Q = \sum_{u\in M}\left[e_{uu} - \left(\sum_{v\in M} e_{uv}\right)^2\right] \tag{8}$$

where M is a set of non-overlapping modules that the network can be completely divided into, and e_{uv} is determined by the ratio of the number of connections that link the nodes in module u with the nodes in module v to the total number of the connections in the network.

Global efficiency of a network is inversely related to the path length of the network and correlates with parallel information transfer, which is defined as

$$E = \frac{1}{n}\sum_{i\in N} E_i = \frac{1}{n}\sum_{i\in N}\frac{\sum_{j\in N, j\neq i} d_{ij}^{-1}}{n-1} \tag{9}$$

where E_i is the efficiency of node i.

ANOVA statistical test

The group differences in the RAPM score and the behavioral performances on the deductive reasoning task were tested by the one-way analysis of variance (ANOVA). The ANOVA was also performed to test the group difference in the fm theta that was derived from the averaged theta-band spectral responses of electrodes FZ and FCZ. In order to discover the significant topological changes occurred in the functional reorganization process of the network, the results of graph-theoretical analysis were statistically tested by the ANOVA between the task and baseline periods. Similar tests were also conducted between the two groups in the task period to find the group differences in the reasoning-related network topology.

Correlation analysis and between-groups discrimination

It has previously been shown that fm theta oscillation emerges in various mental tasks, with the amplitude corresponding to individual effortful cognitive processing (Turken et al., 1999; Onton et al., 2005; Doppelmayr et al., 2008), such as attention concentration, working memory load, self-control, internal timing, and assessment of reward (Nakashima and Sato, 1992; Asada et al., 1999; Jensen and Tesche, 2002; Mizuhara et al., 2004). Importantly, the appearance of event-related fm theta might be a reflection of alternative activities in the medial PFC and AC (Asada et al., 1999). In a nonstationary functional network, individual cognitive effort is also associated with the execution of topological reorganization and even related to the intension of workspace configuration, that is, stronger integration would be driven by higher effortful executive processing and thus more changes in complex network metrics might be measured (Kitzbichler et al., 2011).

According to the above assumption, single-trial correlation analyses were performed to estimate the coordination relationship between mental effort and topological shift related to workspace formation of the theta network. Firstly, the association between the fm theta activated in the reasoning process and the intension of task-related workspace configuration was estimated by the correlation analyses, in which the averaged PSDs from electrodes FZ and FCZ and each global network parameter (i.e., modularity, characteristic path length and global efficiency) in the task period were used as analytical variables. Secondly, by removing the relevant activities in the baseline resting state, the variables used in the correlation analyses were the absolute change rate of averaged PSDs in the two electrodes and the task-related change rate of each global network parameter, which is quantified by

$$R = [(W_T - W_R)/W_R] \times 100\% \tag{10}$$

where W_T and W_R represent the network measures in the task period and pre-stimulus interval, respectively.

Pearson's correlation coefficient r_{PW} can quantify the degree of correlation between two variables, which is defined as

$$\begin{aligned} r_{PW} &= \frac{\sum_{i=1}^{n}(P_i - \overline{P})(W_i - \overline{W})}{\sqrt{\sum_{i=1}^{n}(P_i - \overline{P})^2 \cdot \sum_{i=1}^{n}(W_i - \overline{W})^2}} \\ &= \frac{n\sum_{i=1}^{n}P_iW_i - \sum_{i=1}^{n}P_i\sum_{i=1}^{n}W_i}{\sqrt{n\sum_{i=1}^{n}P_i^2 - (\sum_{i=1}^{n}P_i)^2}\cdot\sqrt{n\sum_{i=1}^{n}W_i^2 - (\sum_{i=1}^{n}W_i)^2}} \end{aligned} \tag{11}$$

where P indicates a variable related to the fm theta, W represents the other selected network measure, and n is the sample size (the number of trials). The range of r_{PW} is within $[-1, 1]$, based on which the correlationship is usually divided into

$$\begin{cases} \text{complete correlation} & |r_{PW}| = 1; \\ \text{high correlation} & 0.8 \le |r_{PW}| < 1; \\ \text{significant correlation} & 0.5 \le |r_{PW}| < 0.8; \\ \text{slight correlation} & 0.3 \le |r_{PW}| < 0.5; \\ \text{non-correlation} & 0 \le |r_{PW}| < 0.3; \end{cases}$$

Besides being used as the variables in the correlation analyses, P and W can also represent the features extracted from the trial samples to distinguish the neural characteristics of the two groups of subjects. Thus, in the two-dimensional feature space composed of trial samples labeled with "gifted" and "non-gifted", P and W can be further viewed as the input feature vectors, i.e., (P_i, W_i), where P_i and W_i represent the features extracted from the ith trial, and $i = 1, \ldots, n$. In each feature space, linear discriminant analysis (LDA) (Webb, 2003) with a 10-fold cross validation was employed to conduct binary-classification of the trial samples for

identifying the giftedness-related effortful performance and dynamic integration of the functional network in response to the cognitive demands of the reasoning task.

RESULTS

Behavioral performance and group difference

In the RAPM psychometric test, the math-gifted subjects have obtained significantly higher scores than the control subjects ($p = 0.0000$). In performing the deductive reasoning task, significant group difference can be observed in the reaction time of correct responses, in which the math-gifted group has significantly shorter reaction time than the control group ($p = 0.0036$). Meanwhile, the math-gifted adolescents have outperformed the control subjects in average response accuracy (Table 1).

Temporal coordination between fm theta response and long-distance coherence

As shown in the ERSP diagrams of electrode FZ (Fig. 3A), the deductive reasoning task has elicited the staged spectral response in the theta frequency band. Corresponding to the time interval of ERSP, the theta-band phase synchronizations between distant electrodes increase in the three stages of the reasoning task, including the enhanced intrahemispheric ERPCOH between frontal-occipitoparietal electrodes (Fig. 3B) and interhemispheric ERPCOH between bilaterally frontal inferior electrodes (Fig. 3C). As compared with the control group, the math-gifted subjects show higher response strength of ERSP in electrode FZ and more significant long-distance intrahemispheric ERPCOH and interhemispheric ERPCOH in the theta frequency band (Fig. 3).

In the global distribution of the theta-band response activated by the three stages of the deduction task, more coherent theta-band power change in frontal-sensorimotor-parietal cortical regions is presented at the stages of premise integration and conclusion validation (Fig. 4), which can be associated with the higher synchronized activity in the fronto-parietal network. Particularly, the brain maps from the math-gifted group present higher global theta oscillations, especially at the frontal midline locations.

In the ANOVA tests for the group differences in the fm theta, significantly higher average PSDs in frontal midline electrodes are found in the math-gifted group in both resting and cognitive states ($p = 0.0000$). From the resting state to task period, the math-gifted subjects also show greater task-related change rate of fm theta than the control group ($p = 0.0327$) (Table 2).

Table 1. Behavioral performance and group difference: mean ± standard deviation; *p*-significance level of the ANOVA

	Math-gifted group	Control group	*p*
RAPM score	33.6 ± 0.8	23.6 ± 4.4	0.0000
Reaction time	831 ± 536 ms	994 ± 655 ms	0.0036
Response accuracy	75.14 ± 12.58%	68.20 ± 15.29%	–

Reasoning-related workspace configuration of the theta network

Significant network reorganization related to workspace formation. By the Louvain method of functional community detection (Blondel et al., 2008), the basic architecture of the resting-state theta network is primarily divided into prefrontal, bilateral frontal/frontotemporal, parietal, occipital and frontoparietal modules. Especially

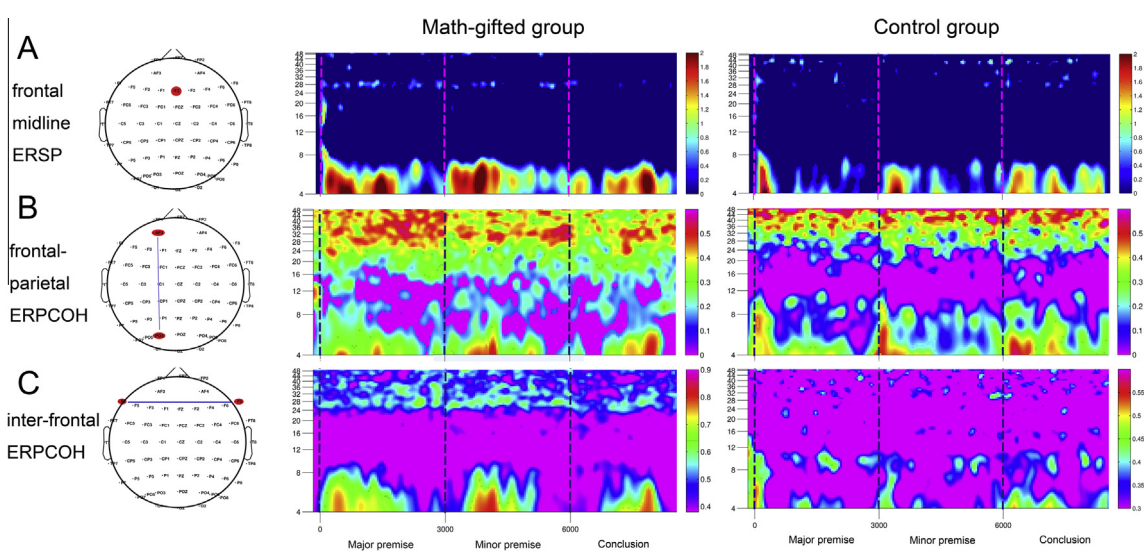


Fig. 3. EEG channel locations and time–frequency analyses for group-based trials: (A) ERSP of electrode FZ located at frontal midline brain area. (B) ERPCOH between electrode pair AF_3–PO_3 at frontal pole-parietooccipital cortex. (C) ERPCOH between electrode pair F1–F8 located at bilateral inferior frontal cortices. The left and right time–frequency diagrams are from the math-gifted group and the control group respectively. In each diagram, horizontal axis is a time-course of the three-stage deductive reasoning task, and the vertical logarithmic axis plots frequency ranging from 4 to 50 Hz.

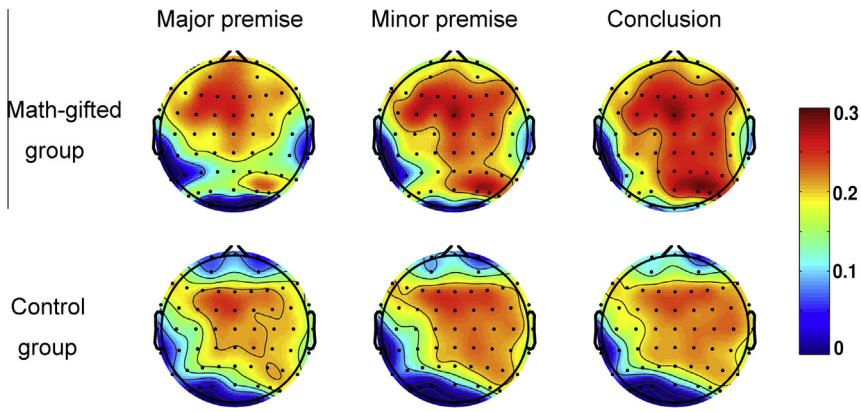


Fig. 4. Brain electric activity mapping of the theta-band absolute power changes during the three stages of the deductive reasoning task: from left to right: the brain maps are derived from the three stages of the task, i.e., major premise, minor premise and conclusion, respectively. From top to bottom: the brain maps are from the math-gifted group and the control group respectively.

Table 2. Fm theta and group difference: mean ± standard deviation; *p*-significance level of the ANOVA

	Math-gifted group	Control group	*p*
Resting state	5.04 ± 1.68	3.79 ± 1.26	0.0000
Task-related change rate	0.34 ± 0.22	0.29 ± 0.20	0.0327
Task period	6.68 ± 1.70	4.58 ± 1.49	0.0000

in the time intervals of premise integration and conclusion validation, more intensive synchronizations across anatomically separated brain areas have increased from the resting state (Fig. 5A). After task-related reconfiguration of connections, the major changes in functional partition of the theta network architecture are found in the integration of some nodes in prefrontal, right frontal and temporal cortical regions, and in the integration of some nodes in parietal and occipital cortical regions, which can be associated with the integrative functions of temporo-frontal and occipito-parietal systems in cognitive processing. Importantly, the reasoning task also induces the integration of some nodes in frontal and parietal regions through the increased long-distance fronto-parietal connections, which leads to an expanded fronto-parietal module in the theta network.

Based on the functional integration by the long-range synchronizations, the network measures of the two groups show significantly decreased modularity and characteristic path length, and significantly increased global efficiency from the resting state to the task period ($p = 0.0000$) (Fig. 6), which indicate that a less modular but more globally synchronized and parallel transferred network configuration pattern has been formed in the reasoning process.

The statistical tests for the individual nodes in the network present the changes of local configurations in the topological reorganization process (Fig. 7). The increase of the node degree is primarily focused on the frontal–sensorimotor–superior parietal cortical regions, accompanied with simultaneously decreased node degree distributed in temporal and inferior parietal and occipital regions (Fig. 7A). Because of the adjustment of node degree distribution, the simultaneously increased within-module degree *z*-score also emerges in the fronto-parietal cortical area, which correlates with the increased interconnectivity of fronto-parietal network in the cognitive process (Fig. 7B). Corresponding to the analytical results of the global network topology, the individual nodes with diminished clustering coefficient and local efficiency are extensively distributed in the multiple cortical regions (Fig. 7C, D), which demonstrates that the local modularity of the network has been greatly weakened in the reasoning process, following the globally synchronized and less modular network configuration. According to these changes of the key graph-theoretical parameters related to workspace formation in the complex network organization, the global neuronal workspace in theta frequency of the brain activity has been driven by the subjects for supporting higher cognitive functions of the reasoning task, by means of long-range recruitment of original loosely connected functional modules.

Group difference in network reorganization and workspace configuration. In the math-gifted group, it can be found that more node pairs with increased synchronizations in the fronto-parietal network have been formed in the reasoning task (Fig. 5A). Thus, the math-gifted group shows greater topological shift of the theta network from the resting state to the reasoning process than the control group, which is demonstrated by higher absolute change rates of characteristic path length (mean ± SD: −0.15 ± 0.08 vs. −0.13 ± 0.09; $p = 0.0179$), modularity (mean ± SD: −0.16 ± 0.06 vs. −0.14 ± 0.07; $p = 0.0020$) and global efficiency (mean ± SD: 0.09 ± 0.07 vs. 0.07 ± 0.11; $p = 0.0287$) (Fig. 8B). Meanwhile, corresponding to the changes in global network measures, the math-gifted subjects show more significant changes in nodal measures, and quantitatively more cortical vertices significantly involved in the network reorganization process (Fig. 7A–D). From the above comparisons at both global and nodal levels of the theta network, the analytical results suggest the highly adaptive functional self-organization of neuronal assemblies and greater transition capacity from locally

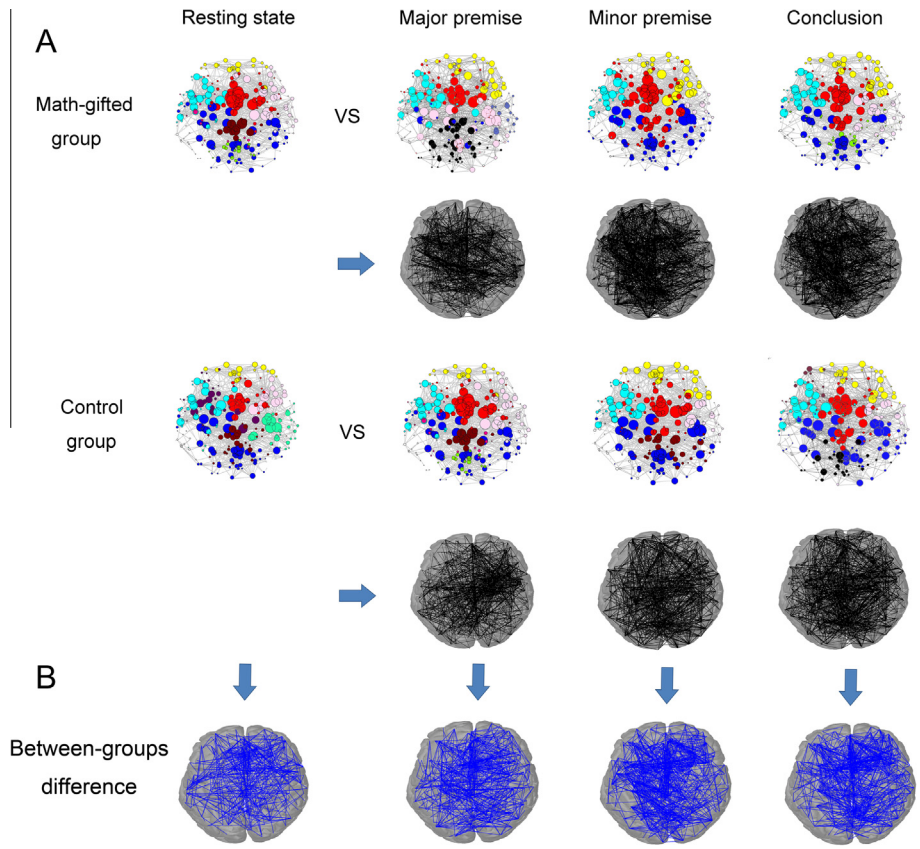


Fig. 5. Functional partitions and differences in synchronized node pairs of the theta networks in the resting state and three stages of the deductive reasoning task: (A) adaptive network reconfiguration process. In each network topology, a node represents a cortical vertex, with the color indicating its membership of a functional module. The size of a node is proportional to "degree" of the node. The black edges represent the increased synchronized node pairs in the three stages of the deductive reasoning task with respect to the resting state. (B) Between-groups comparison. The blue edges indicate the increased synchronized node pairs in the math-gifted group compared to the control group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

modular community structure to more integrated network architecture in the math-gifted brain.

In the reasoning process, the ANOVA tests indicate that the theta network of the math-gifted subjects works in a topological pattern with stronger workspace configurations than the control subjects, through less modularity (mean ± SD: 0.47 ± 0.05 vs. 0.49 ± 0.05; $p = 0.0000$), shorter path length (mean ± SD: 2.65 ± 0.20 vs. 2.77 ± 0.29; $p = 0.0000$) and higher global efficiency of parallel information transfer (mean ± SD: 0.35 ± 0.02 vs. 0.33 ± 0.04; $p = 0.0000$) (Fig. 8C). At the nodal level of the reasoning-related theta network, the nodes with increased local configuration measures in the math-gifted group are mainly distributed on the fronto-parietal brain regions (Fig. 9). Especially in the anterior cortical vertices in medial and right frontal modules, the math-gifted group has significantly higher node degree, within-module degree *z*-score, clustering coefficient and efficiency than the control subjects (Table 3).

Although the increased global integration in the task-related theta network of the math-gifted brain would weaken the local cluster of nodes, the relatively higher clustering coefficient and local efficiency in the medial and right frontal modules indicate that the math-gifted adolescents still have maintained the local cliques and the local information transfers better than the control subjects in the right-lateral anterior brain area (Fig. 9C, D). The good performance can be attributed to the higher node degree distribution (i.e., higher connection density) and better interconnectivity of the nodes in their anterior sub-network (Fig. 9A, B).

Correlationship between fm theta and workspace configuration of theta network

During the reasoning process, there are correlationships between the task-related fm theta activation and the workspace configurations of the theta network. As shown in the scattergrams composed of all the trial samples, the PSD of the fm theta shows slight negative correlation with the modularity and the characteristic path length, and slight positive correlation with the global efficiency of the theta network, respectively (correlation coefficient $|r_{PW}| > 0.3$). In the single-trial discrimination for group-based samples, the math-gifted adolescents can be indentified as having higher fm theta activation and stronger workspace pattern of the reasoning-related theta network (the accuracies are higher than 0.74) (Fig. 10A).

Besides, the change rates of the global network measures toward workspace configuration can be

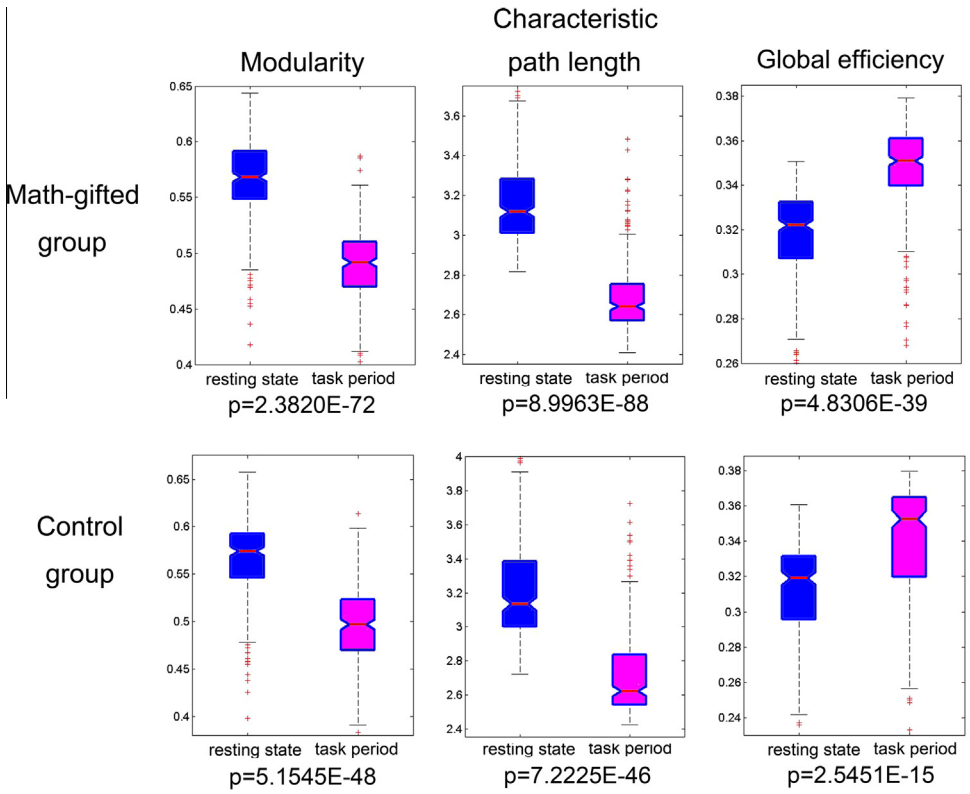


Fig. 6. Statistical boxplots of ANOVA tests between global network parameters in the resting state and the task period: from left to right: modularity, characteristic path length and global efficiency of the theta network.

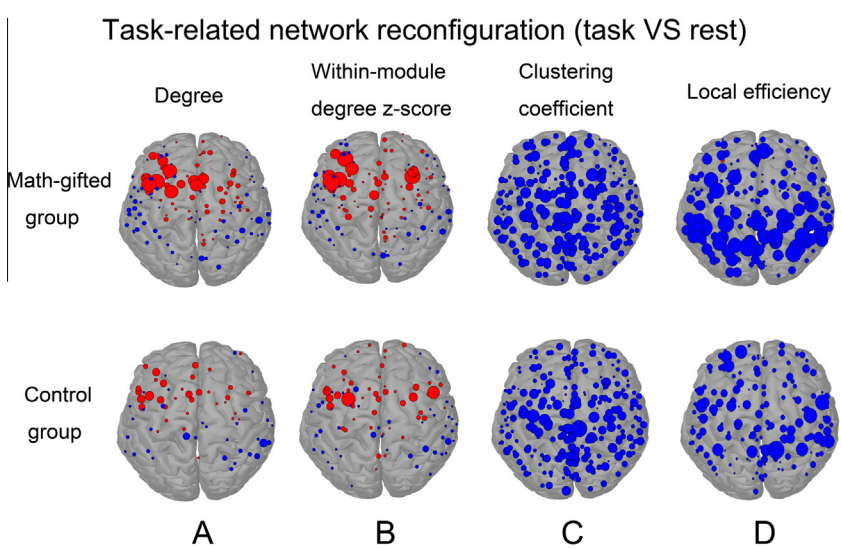


Fig. 7. Individual nodes with significant changes in theta network reorganization (task vs. rest with significance level $p < 0.05$): from left to right: nodes with significant change in (A) degree, (B) within-module degree *z*-score, (C) clustering coefficient and (D) local efficiency. The red circles represent nodes with significantly increased parameter values in the task period as compared with the resting state, and the blue circles refer to significant decrease. The size of a node is proportional to its log*p* value in the ANOVA test with the null hypothesis that the task-induced change is zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

coordinated with the change rate of the fm theta power in the trial samples, that is, the reducing modularity and path length and the increasing global efficiency follow the increasing task-induced fm theta response (Fig. 10B).

DISCUSSIONS

Theta-band activity involved in the deductive reasoning task. Unlike the mental rotation and RAPM tasks, the three-stage deductive reasoning task strongly requires

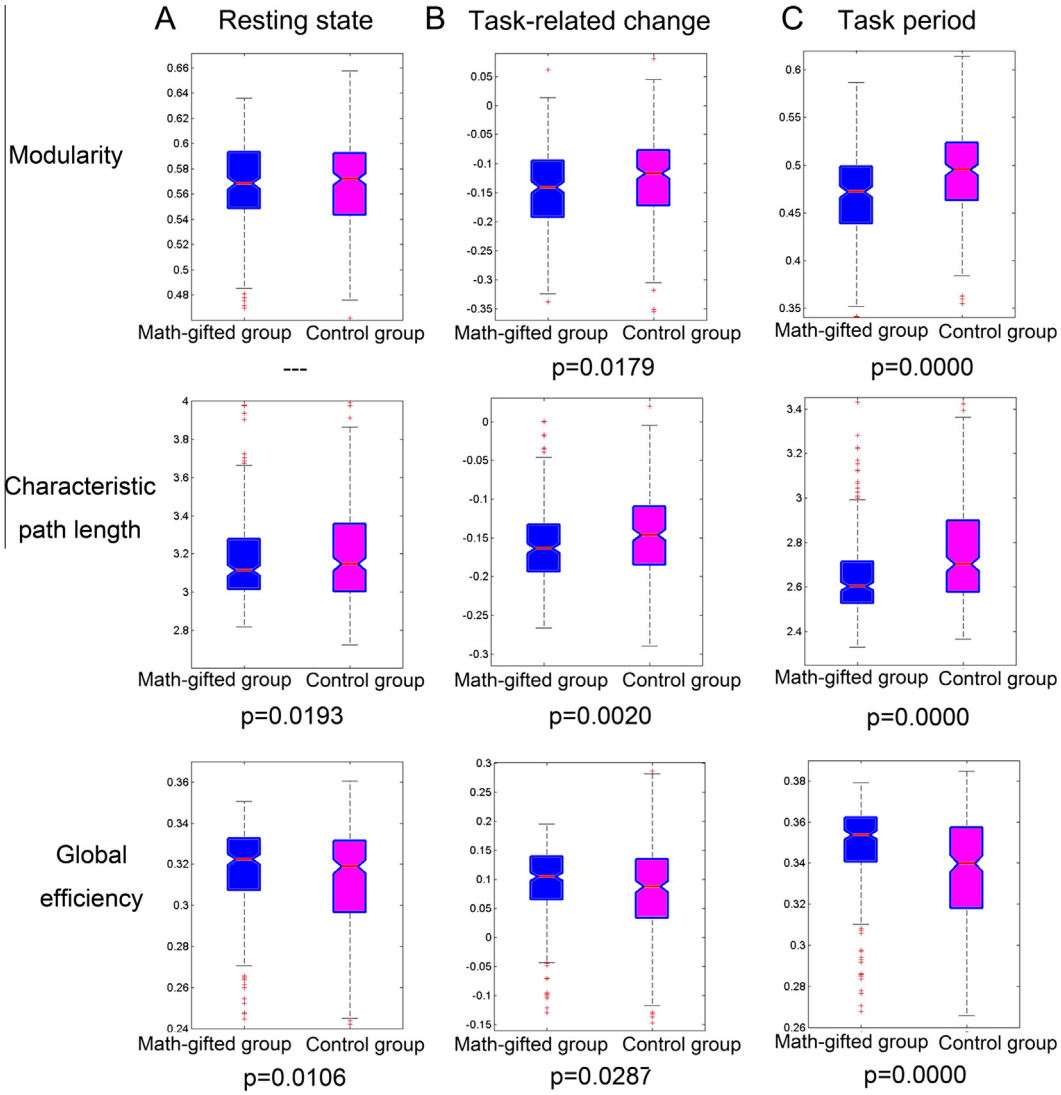


Fig. 8. Statistical boxplots of between-groups ANOVA tests for global network parameters: from top to bottom: modularity, characteristic path length and global efficiency. (A) Global parameters of the resting-state network. (B) Task-related change rate of global parameters. (C) Global parameters in the task-related network.

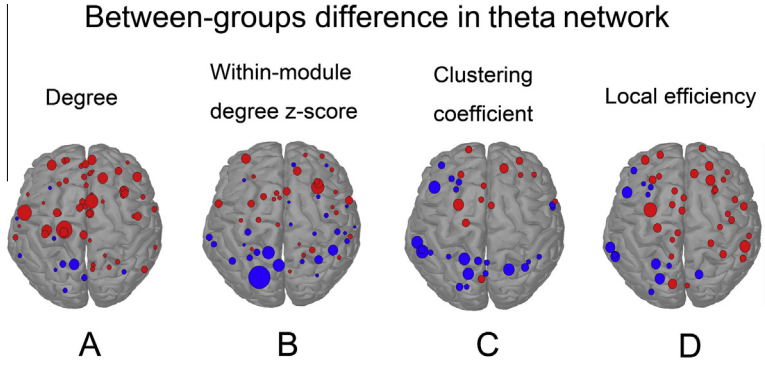


Fig. 9. Individual nodes with significant group differences in the theta network topology in the task period (math-gifted group vs. control group with significance level $p < 0.05$): from left to right: nodes with significant group differences in (A) degree, (B) within-module degree z-score, (C) clustering coefficient and (D) local efficiency. The red circles represent nodes with significantly increased parameter values in the math-gifted group as compared with the control group, and the blue circles refer to a significant decrease. The size of a node is proportional to its $\log p$ value in the ANOVA test with the null hypothesis that the group difference is zero. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

individual attention concentration for remaining on task, continuous processing of premise information, information maintenance and memory retention in the working memory system, and response comparison between the displayed conclusion and the inference drawn from premises (Knauff et al., 2002; Fangmeier et al., 2006). It depends on individual reasoning ability and effortful cognitive processing through interconnected frontal and parietal cortical circuitry (Blair, 2007).

The time–frequency analyses discover that, theta-band neural activity is significantly involved in the deductive reasoning process, through the staged spectral response and the emergence of long-distance phase coherence. As an important building-block of functional signaling, theta oscillation is highly involved in cognitive load and memory performances (Klimesch, 1999; Sauseng et al., 2005). Thus, the significant theta response recorded in this experiment can be associated with mental reasoning-induced workload required in different stages of the task. Moreover, the theta fronto-parietal coherence can be viewed as a reflection of central executive functions of working memory load and mirrors the activation of the fronto-parietal network (Sauseng et al., 2005). For the math-gifted adolescents, while responding to more intensive fm theta response regarding higher mental effort of workload manipulations (Itthipuripat et al., 2013), the enhanced theta long-range coherence in the fronto-parietal network reflects their more expanded memory capacity in encoding and maintaining information via the highly integrated fronto-parietal cortical circuitry in the reasoning process.

Table 3. Local configuration parameters of reasoning-related theta network and group difference: mean ± standard deviation; *p*-significance level of the ANOVA

	Math-gifted group	Control group	*p*
A. Mean value resulted from right frontal nodes			
Degree	11.53 ± 3.73	9.53 ± 4.25	0.0000
With-module degree *z*-score	0.25 ± 0.48	−0.03 ± 0.40	0.0000
Clustering coefficient	0.37 ± 0.10	0.31 ± 0.12	0.0000
Local efficiency	0.57 ± 0.13	0.48 ± 0.16	0.0000
B. Mean value resulted from medial frontal nodes			
Degree	11.05 ± 2.42	9.64 ± 2.74	0.0000
With-module degree *z*-score	0.62 ± 0.39	0.34 ± 0.43	0.0000
Clustering coefficient	0.38 ± 0.09	0.34 ± 0.09	0.0000
Local efficiency	0.57 ± 0.15	0.47 ± 0.16	0.0000

Functional network reorganization and workspace pattern of network configuration. By means of increased long-range synchronizations, the math-gifted adolescents exhibit greater shift of topological organization to "workspace" configuration than the control subjects, which is manifested as more significant changes of network measures toward a shorter global path, higher global efficiency, less modularity, less local clustering, and better-interconnected fronto-parietal network architecture. This tendency of network reorganization and the enhanced topological workspace configuration in the math-gifted brain can better facilitate task-related functional interaction among discrete brain regions and promote the level of global synchronization

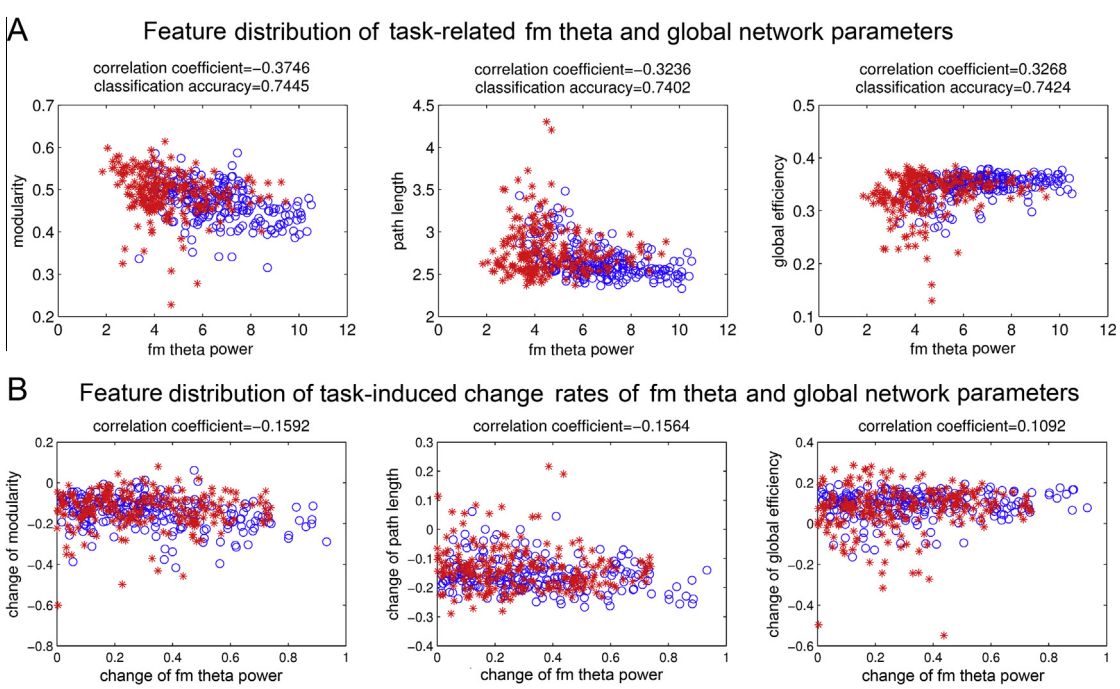


Fig. 10. Scattergrams of trial samples with fm theta power and global topological parameters of the theta network (modularity, characteristic path length, or global efficiency) as features: (A) fm theta power and global parameters of the theta network in the task period; (B) task-related change rates of fm theta power and global parameters. The blue circles and the red asterisks represent the trial samples from the math-gifted subjects and the control subjects respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in the reasoning task, which benefit efficient information storage, processing and transfer in the crucial fronto-parietal network.

In addition, the between-group comparisons find that the math-gifted adolescents have increased local measures in the nodes of the medial and right frontal modules. The previous studies have emphasized the importance of the right frontal lobe in mathematical problem-solving, as it is dominant for geometric and visual space analyses and also highly involved in mathematical creativity (O'Boyle et al., 2005; Prescott et al., 2010). The better local configurations of the right-lateral frontal nodes in the math-gifted brain could support more efficient local information exchange and even facilitate the unity of various thoughts arising from mathematical reasoning and creative thinking processes.

Coordination relationship between mental effort and workspace formation. In the time–frequency diagrams, the emergence of long-range synchronizations can be coordinated in the time interval of fm theta response while executing different stages of the reasoning task, which is in good agreement with the results reported by the study using simultaneously recorded EEG and functional magnetic resonance imaging (fMRI) (Mizuhara et al., 2004). Our study discovers that there are slight correlations between fm theta power and global network measures in terms of workspace configuration, in which the math-gifted group can be characterized by an enhanced state of effort-driven topological shift toward workspace formation. The findings can support Kitzbichler et al.'s (2011) research conclusion that higher cognitive load (mental effort) can drive greater functional reorganization and thus facilitate stronger workspace formation of human brain functional network.

The global neuronal workspace theory suggests that, through integrative function of the neural system, the messages from the global workspace are broadcast to all the unconscious processors to recruit their participation for solving novel problems (Baars, 2005). From the findings in our study, with the well-developed anterior neural system (O'Boyle et al., 2005; Desco et al., 2011), the math-gifted brain could be assumed to have a superior central executive processor that can facilitate/mobilize greater recruitment of the originally competing processors distributed in discrete brain regions to cooperatively participate in the complex cognitive process.

It should be noted that previous neuroimaging studies have suggested different neurobiological substrates unique to giftedness in mathematics or common to intelligence. In cognitive processing, the highly activated anterior brain areas including the PFC, frontal lobe and AC can be connected to general fluid intelligence of math-gifted adolescents (O'Boyle et al., 2005). In our study, the greater effort-driven network reorganization seems to be attributed to the nature of high intelligence of the math-gifted adolescents irrespective of their specific aptitude in mathematics. This suggestion is rather consistent with the previous research findings on intelligent brains that bright individuals often devote more effortful cortical resources while there are quite strong cognitive requirements (Neubauer and Fink, 2009).

Additionally, there are original group differences in the resting-state theta cortical network organization. Compared with the control group, the math-gifted adolescents have shorter characteristic path length (mean ± SD: 3.18 ± 0.24 vs. 3.25 ± 0.38; $p = 0.0193$) and higher global efficiency (mean ± SD: 0.32 ± 0.02 vs. 0.31 ± 0.03; $p = 0.0106$) (Fig. 8A), which indicates higher global synchronization activity under resting condition without any task information processing. From the between-groups comparison in the resting-state network (Fig. 5B), it can be found that the differences in the global network measures are primarily resulted from the increased fronto-parietal synchronizations in the math-gifted brain. Complex brain network research has suggested that resting-state functional connectivity on multiple time scales is shaped by underlying anatomical architecture of the cerebral cortex (Honey et al., 2007). A recent diffusion tensor imaging (DTI) study has demonstrated the increased anatomical connectivity in fronto-parietal association tracts in the math-gifted brain, and further found it underlies subjects' specific mathematical ability independent of their intelligence scores (Navas-Sanchez et al., 2013). In conjunction with the dynamic reconfiguration of the functional network, the topological basis of the resting-state cortical network probably underlying greater mathematical ability should also be an important part of the stronger emergence of workspace configurations in the reasoning-related functional network.

CONCLUSIONS

By constructing the theta-band cortical source networks under resting and task conditions, our EEG study has confirmed the greater topological reorganization of the brain functional network in the math-gifted adolescents in response to the cognitive demands from a deductive reasoning task. Coordinated with stronger cognitive effort invested in the reasoning process, the math-gifted adolescents exhibited an enhanced task-related global workspace at the level of global network organization than the average-ability control subjects. With a better-integrated resting-state cortical network, the higher capacity of the math-gifted brain in the dynamic transition of topological organization from modular subsystems to global workspace can further promote the level of timely information processing of the fronto-parietal network. Additionally, our complex network analysis indicates that the dynamically reconfigured network architecture in the math-gifted brain enhances both global and local information processing in reasoning processes, which can be viewed as the neural correlations with the superior problem-solving ability of the math-gifted adolescents in logical reasoning.

Acknowledgments—This work was supported in part by the National Natural Science Foundation of China under Grant 31130025, the National Basic Research Program of China under Grant 2015CB351704, and the National Natural Science Foundation of China under Grant 61375118, and the Program for New Century Excellent Talents in University of China under Grant NCET-12-0115. The authors would like to thank the anonymous reviewers for their thoughtful comments and suggestions.

REFERENCES

Asada H, Fukuda Y, Tsunoda S, Yamaguchi M, Tonoike M (1999) Frontal midline theta rhythms reflect alternative activation of prefrontal cortex and anterior cingulate cortex in humans. Neurosci Lett 274:29–32.

Baars BJ (2005) Global workspace theory of consciousness: toward a cognitive neuroscience of human experience. Prog Brain Res 150:45–53.

Banfield T (2005) Ability grouping for mathematically gifted adolescent boys. Int Educ J 6:141–149.

Basar E, Schurmanna U, Sakowitz O (2001) The selectively distributed theta system: functions. Int J Psychophysiol 39:197–212.

Bassett DS, Meyer-Lindenberg A, Achard S, Duke T, Bullmore E (2006) Adaptive reconfiguration of fractal small-world human brain functional networks. Proc Natl Acad Sci U S A 103:19518–19523.

Blair C (2007) Inherent limits on the identification of a neural basis for general intelligence. Behav Brain Sci 30:154–155.

Blondel VD, Guillaume JL, Lambiotte R, Lefebvre E (2008) Fast unfolding of communities in large networks. J Stat Mech 2008:P10008.

Bullmore E, Sporns O (2009) Complex brain networks: graph theoretical analysis of structural and functional systems. Nat Rev Neurosci 10:186–198.

Cooper-Martin E (1994) Measures of cognitive effort. Mark Lett 5:43–56.

Dehaene S, Changeux JP (2005) Ongoing spontaneous activity controls access to consciousness: a neuronal model for in attentional blindness. PLoS Biol 3:e141.

Dehaene S, Kerszberg M, Changeux JP (1998) A neuronal model of a global workspace in effortful cognitive tasks. Proc Natl Acad Sci U S A 95:14529–14534.

Dehaene S, Naccache L (2001) Towards a cognitive neuroscience of consciousness: basic evidence and a workspace framework. Cognition 79:1–37.

Delorme A, Makeig S (2004) EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. J Neurosci Methods 134:9–21.

Desco M, Navas-Sanchez FJ, Sanchez-Gonzalez J, Reig S, Robles O, Franco C, Guzman-De-Villoria JA, Garcia-Barreno P, Arango C (2011) Mathematically gifted adolescents use more extensive and more bilateral areas of the fronto-parietal network than controls during executive functioning and fluid reasoning tasks. Neuroimage 57:281–292.

Doppelmayr M, Finkenzeller T, Sauseng P (2008) Frontal midline theta in the pre-shot phase of rifle shooting: differences between experts and novices. Neuropsychologia 46:1463–1467.

Erdös P, Rényi A (1961) On the strength of connectedness of a random graph. Acta Math Hungar 12:261–267.

Fangmeier T, Knauff M, Ruff CC, Sloutsky V (2006) FMRI evidence for a three-stage model of deductive reasoning. J Cogn Neurosci 18:320–334.

Gagne F (1985) Giftedness and talent: reexamining a reexamination of the definitions. Gift Child Q 29:103–112.

Gardner H (2006). In: Gardner H, editor. Multiple intelligence: new horizons. New York: Basic Books.

Goel V (2007) Anatomy of deductive reasoning. Trends Cogn Sci 11:435–441.

Gramfort A, Papadopoulo T, Olivi E, Clerc M (2010) OpenMEEG: opensource software for quasistatic bioelectromagnetics. Biomed Eng Online 9:45.

Honey CJ, Kötter R, Breakspear M, Sporns O (2007) Network structure of cerebral cortex shapes functional connectivity on multiple time scales. Proc Natl Acad Sci U S A 104:10240–10245.

Itthipuripat S, Wessel JR, Aron AR (2013) Frontal theta is a signature of successful working memory manipulation. Exp Brain Res 224:255–262.

Jensen O, Tesche CD (2002) Frontal theta activity in humans increases with memory load in a working memory task. Eur J Neurosci 15:1395–1399.

Jung RE, Haier RJ (2007) The parieto-frontal integration theory (P-FIT) of intelligence: converging neuroimaging evidence. Behav Brain Sci 30:135–154.

Kitzbichler MG, Henson RN, Smith ML, Nathan PJ, Bullmore ET (2011) Cognitive effort drives workspace configuration of human brain functional networks. J Neurosci 31:8259–8270.

Klados MA, Kanatsouli K, Antoniou I, Babiloni F, Tsirka V, Bamidis PD, Micheloyannis S (2013) A graph theoretical approach to study the organization of the cortical networks during different mathematical tasks. PLoS One 8:e71800.

Klimesch W (1999) EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. Brain Res Rev 29:169–195.

Knauff M, Mulack T, Kassubek J, Salih HR, Greenlee MW (2002) Spatial imagery in deductive reasoning: a functional MRI study. Cogn Brain Res 13:203–212.

Krutetskii VA (1976) The psychology of mathematical abilities in school children. Chicago, IL: University of Chicago Press.

Lachaux JP, Rodriguez E, Martinerie J, Varela FJ (1999) Measuring phase synchrony in brain signals. Hum Brain Mapp 8:194–208.

Langer N, Pedroni A, Gianotti LR, Hanggi J, Knoch D, Jancke L (2012) Functional brain network efficiency predicts intelligence. Hum Brain Mapp 33:1393–1406.

Markov NT, Ercsey-Ravasz M, Lamy C, Gomes ARR, Magrou L, Misery P, Giroud P, Barone P, Dehay C, Toroczkai Z, Knoblauch K, Van Essen DC, Kennedy H (2013) The role of long-range connections on the specificity of the macaque interareal cortical network. Proc Natl Acad Sci U S A 110:5187–5192.

Mizuhara H, Wang LQ, Kobayashi K, Yamaguchi Y (2004) A long-range cortical network emerging with theta oscillation in a mental task. Neuroreport 15:1233–1238.

Mizuhara H, Yamaguchi Y (2007) Human cortical circuits for central executive function emerge by theta phase synchronization. Neuroimage 36:232–244.

Monti MM, Osherson DN, Martinez MJ, Parsons LM (2007) Functional neuroanatomy of deductive inference: a language-independent distributed network. Neuroimage 37:1005–1016.

Nakashima K, Sato H (1992) The effects of various mental tasks on appearance of frontal midline theta activity in EEG. J Hum Ergol 21:201.

Navas-Sanchez FJ, Aleman-Gomez Y, Sanchez-Gonzalez J, Guzman-De-Villoria JA, Franco C, Robles O, Arango C, Desco M (2013) White matter microstructure correlates of mathematical giftedness and intelligence quotient. Hum Brain Mapp 35:2619–2631.

Neubauer AC, Fink A (2009) Intelligence and neural efficiency. Neurosci Biobehav Rev 33:1004–1023.

O'Boyle MW, Cunnington R, Silk TJ, Vaughan D, Jackson G, Syngeniotis A, Egan GF (2005) Mathematically gifted male adolescents activate a unique brain network during mental rotation. Cogn Brain Res 25:583–587.

Onton J, Delorme A, Makeig S (2005) Frontal midline EEG dynamics during working memory. Neuroimage 27:341–356.

Pfurtscheller G, Silva FHLD (1999) Event-related EEG/MEG synchronization and desynchronization: basic principles. Clin Neurophysiol 10:1842–1857.

Prado J, Chadha A, Booth JR (2011) The brain network for deductive reasoning: a quantitative meta-analysis of 28 neuroimaging studies. J Cogn Neurosci 23:3483–3497.

Prescott J, Gavrilescu M, Cunnington R, O'Boyle MW, Egan GF (2010) Enhanced brain connectivity in math-gifted adolescents: an fMRI study using mental rotation. Cogn Neurosci 1:277–288.

Rodriguez-Moreno D, Hirsch J (2009) The dynamics of deductive reasoning: an fMRI investigation. Neuropsychologia 47:949–961.

Rubinov M, Sporns O (2010) Complex network measures of brain connectivity: uses and interpretations. Neuroimage 52: 1059–1069.

Sakkalis V (2011) Review of advanced techniques for the estimation of brain connectivity measured with EEG/MEG. Comput Biol Med 41:1110–1117.

Sauseng P, Klimesch W, Schabus M, Doppelmayr M (2005) Fronto-parietal EEG coherence in theta and upper alpha reflect central executive functions of working memory. Int J Psychophysiol 57:97–103.

Sauseng P, Griesmayr B, Freunberger R, Klimesch W (2010) Control mechanisms in working memory: a possible function of EEG theta oscillations. Neurosci Biobehav Rev 34:1015–1022.

Tadel F, Baillet S, Mosher JC, Pantazis D, Leahy RM (2011) Brainstorm: a user-friendly application for MEG/EEG analysis. Comput Intell Neurosci 2011:8.

Turken AN, Swick D, Martinez VA (1999) Spectral changes in the EEG associated with error processing. J Cogn Neurosci:61.

Webb AR (2003) Statistical pattern recognition. Hoboken, NJ: John Wiley & Sons.

Wharton CM, Grafman J (1998) Deductive reasoning and the brain. Trends Cogn Sci 2:54–59.

(Accepted 23 December 2014)
(Available online 13 January 2015)